

File permissions in Linux

Project description

The research team needs to modify the permissions for certain files and directories inside the projects directory. Updating the file permissions is crucial to make the system secure, therefore, the steps I performed for this task are:

Check file and directory details

Command: **ls -la**; provides the output of all files and directories (even the hidden ones) inside the projects directory and their respective file permissions for owners, groups and others. The output showed me that the file `.project_x.txt` is the sole hidden file, while other files were not hidden but the privileges were still misconfigured.

Describe the permissions string

The permissions string is a string of 10 characters:

- **1st** character represents the type of file ("d" stands for directory, while "-" stands for regular files).
- **2nd-4th** characters represent the read, write and execute permissions of the owner of the file respectively (rwx). If some character is substituted with the "-", that means that the user does not have the permission for that action.
- **5th-7th** characters represent the read, write and execute permissions of the group users of the file respectively (rwx). If some character is substituted with the "-", that means that the user does not have the permission for that action.
- **8th-10th** characters represent the read, write and execute permissions of the other users of the file respectively (rwx). If some character is substituted with the "-", that means that the user does not have the permission for that action.

Change file permissions

To change the file permissions we use the `chmod` command in combination with the user group and file permissions that we want to add or take away. We have already discussed the file permissions (r - read, w - write, x - execute), and we also have the (u - user, g - group, o - others) in combination with "+" or "-" sign to give or take away permission. If we want to add read and write permissions to group users we use the `chmod` command like this: **`chmod g+rw filename`**. After the file permissions change it is recommended to look for the permissions table again with **`ls -l(a)`** to check if the permissions were set correctly.

Change file permissions on a hidden file

Changing the file permissions on a hidden file is a very similar process like changing them on normal files except we need to put the “.” symbol before the file name. Since we have only one hidden file (.project_x.txt) that needs to have only read permissions for both owner and the group users we first deleted the write permissions for both of them (**chmod u-w .project_x.txt**, **chmod g-w .project_x.txt**) and after that gave the read permissions to group users (**chmod g+r .project_x.txt**).

Change directory permissions

We have only one directory inside the projects folder - drafts. The owner has all file permissions for the drafts directory, but my organization asked to remove the group execute options for the directory so we have to take the execute permission away from the group. I did it by executing the: **chmod g-x drafts**

Summary

The file permissions were configured in accordance with the organization's needs. For viewing the current permissions we needed to access them through **ls -la** command. Rest of the work was done by the **chmod** command, which helped us change the file permissions of the files and a directory.